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Aerobic oxidative esterification of 5-hydroxymethylfurfural to dimethyl furan-2,5-dicarboxylate by using homogeneous and heterogeneous PdCoBi/C catalysts under atmospheric oxygen†

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The conversion of platform molecule 5-hydroxymethylfurfural (HMF) into many value-added derivatives has attracted significant interest. FDCA and its esters are important derivatives of HMF, which can be used as polyester monomers and pharmaceutical intermediates. In this paper, oxidative esterification of 5-HMF has been carried out by using homogeneous and heterogeneous PdCoBi/C catalysts under atmospheric oxygen. The effect of reaction conditions on product distribution has been studied under both homogeneous and heterogeneous catalytic conditions. The highest yields of oxidative esterification products are obtained at 93% and 96% by using homogeneous and heterogeneous PdCoBi/C catalysts, respectively. The catalysts are characterized by X-ray photoelectron spectroscopy (XPS) and powder X-ray diffraction (XRD). The catalytic system has better compatibility according to the expansion of the substrate. A reaction mechanism is proposed, and recycle experiments are also conducted.

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Introduction

The rapid consumption of fossil fuel resources has caused significant environmental pollution, and it could lead to an energy crisis. Thus, exploring sustainable green energy is very urgent and necessary. Biomass, as the only renewable carbon resource, can be converted into high value-added fuels and chemicals, which can play an important role in solving energy and environmental problems.¹ Biomass-based polymer materials have dual functions of resource conservation and environmental protection. At present, the widely used biomass-based polymer materials are mainly fatty acid-based polymers such as polylactic acid (PLA),^{2a} polyhydroxy fatty acid (PHA),^{2b} polyglycolic acid (PGA),^{2c} and polybutylene glycol succinate (PBS);^{2d} their mechanical properties and heat resistances are significantly lower than those of petroleum-based polymers such as aromatic polyethylene terephthalate (PET) and bisphenol A epoxy resin (Epoxy), which severely limits their applications. 2,5-Furandicarboxylic acid (FDCA) is one of

the potential bio-based platform compounds evaluated by the U.S. Department of Energy; it is considered to be a “sleeping giant” by DuPont and DSM corporations.³ The novel polyethylene-2,5-furandicarboxylate (PEF) compound prepared by polymerization of 2,5-furandicarboxylic acid (FDCA) or its esters and ethylene glycol has excellent material properties. The oxygen isolation rate of PEF is six times higher than that of PET, and the isolation rate of CO₂ and water of PEF is two times higher than that of PET. The glass transition temperature of PEF is 11 degrees higher than that of PET, and the melting temperature of PEF is 40 degrees lower than that of PET.⁴ Similar to PTA, FDCA also has an aromatic ring system and a diacid structure, which is required for the synthesis of the polyester. FDCA is expected to be an important biomass-based polyester material for the synthesis of green, degradable plastics and non-toxic plasticizers.⁵ Therefore, it is of great significance to study the green and efficient synthesis of FDCA and its esters.

According to the different sources of raw materials, the methods of preparation of 2,5-FDCA have been mainly divided into the following categories: (1) preparation using furoic acid through a disproportionation reaction or chloromethylation reaction or CO₂ solid-phase reaction.⁶ (2) Preparation using hexaric acid through an oxidation dehydration reaction.⁷ The raw materials are easily isomerized and carbonized under acidic conditions, which results in low yield; (3) diglycolic acid is used as the raw material to react with methanol and thionyl chloride to obtain diglycolic acid dimethyl ester, which is then

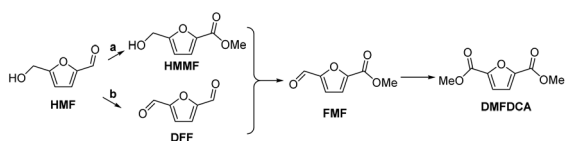
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condensed with trimeric glyoxal dehydrate. However, this method is expensive and not suitable for large-scale industrial production.⁸ (4) Preparation using furan, 2,5-dimethyl furan, 2-methyl-5-acetyl furan, 5-methyl-2-furfural and other furan-based materials through oxidative reactions. Under acidic conditions, the opening of the furan ring is more likely to cause incomplete oxidation, and the synthesis of the raw materials is difficult, which greatly reduces the selectivity of 2,5-FDCA.⁹ (5) Preparation using 5-HMF obtained from the hydrolysis of hexoses. HMF can be further converted into a wide variety of value-added derivatives.¹⁰ The oxidation of HMF is the most studied reaction process at present as the structure of HMF is analogous with that of FDCA, *i.e.*, they both have similar functional group positions.

The traditional pathway for the oxidation of HMF to FDCA involves equivalent oxidizing agents such as nitric acid and^{11a} potassium permanganate;^{11b} however, large quantities of waste salt and a relatively low reaction yield are significant problems that hinder its use in industrial production. In light of this, homogeneous catalysts such as the Co/Mn/Br catalyst^{12a} and TEMPO catalysts^{12b} were developed to improve the reaction conditions. However, some problems still remained such as reactor corrosion and difficulty in catalyst recovery. Therefore, numerous supported metal or metal oxide heterogeneous catalytic systems were developed (*e.g.*, Au,¹³ Pd,¹⁴ Ru,¹⁵ Fe,¹⁶ *etc.*) for the conversion of HMF to FDCA, which resulted in higher selectivity and yields. However, due to the special physical and chemical properties of FDCA, its solubility is poor in water and most organic solvents. Purification by conventional crystallization and rectification has been difficult, which has limited the application scale of FDCA. Furan dicarboxylic acid esters, especially dimethyl furan dicarboxylate (DMFDCA), can be easily separated and purified from the reaction mixture due to its lower boiling point, and it can be used as an ideal substitute for FDCA. Therefore, it is of great significance to study the selective one-step oxidation of 5-HMF to DMFDCA. The oxidative esterification of HMF to DMFDCA proceeds *via* two reaction pathways: (1) the aldehyde group is converted to 5-hydroxymethyl-2-methyl-furoate through oxidative esterification; then, the hydroxyl is oxidized to 5-formyl-2-methyl-furoate; (2) the hydroxyl group is first oxidized to DFF and then, the aldehyde group is esterified to 5-formyl-2-methyl-furoate. Finally, the aldehyde group is oxidized and esterified to obtain DMFDCA (Scheme 1).

In 2008, Christensen *et al.* realized the catalytic oxidation of 5-HMF to DMFDCA in the presence of sodium methoxide at 130 °C and 4 atm *via* an Au/TiO₂ catalyst with a yield of 98%.¹⁷



Scheme 1 The reaction pathways of HMF under oxidative esterification conditions.

Corma *et al.* reported the oxidative esterification of 5-HMF with Au/CeO₂ as a catalyst at 130 °C and 10 atm O₂ to obtain 99% yield.¹⁸ Subsequently, Maella Manzoli developed a catalytic system for the oxidative esterification of furfural and HMF over an Au/ZrO₂ catalyst under 1 atm O₂, which resulted in high yields and good selectivity.¹⁹ In addition to the Au-based catalysts, Fu *et al.* reported that DMFDCA can be prepared *via* a CoxOy-N@C catalyst (100 °C, 10 atm O₂) with 96% yield.²⁰ In 2016, high yield of furan-2,5-dimethylcarboxylate (FDMC, 95.6%) without a basic additive was obtained by using the CoOx-N/C and α -MnO₂ catalytic system (6 atm O₂, 100 °C).²¹ Recently, Tong *et al.* used an MnO₂-Cu catalyst (3 atm O₂, 140 °C), and they found that the selectivity of DMFDCA was only 26.5%.²² Stahl *et al.* reported that alcohols can be oxidized to esters or acids with high selectivity by PdBiTe catalysts.²³ Overall, the selective aerobic oxidative esterification of HMF still requires high temperature or certain pressure. Therefore, it is still a challenge to develop a highly selective catalytic system for the oxidative esterification of 5-HMF to DMFDCA under mild conditions with both normal pressure and low temperature.

Herein, a homogeneous catalytic system and a heterogeneous catalytic system were designed for the selective oxidative esterification of HMF to DMFDCA. The effect of various additives on product distribution was investigated initially. Based on these studies, the reaction conditions such as catalyst loadings, catalyst proportion, alkali types and alkali amounts were screened subsequently. To realize the recyclability of the catalytic system, a heterogeneous catalyst was prepared. Meanwhile, characterizations using XRD and XPS were carried out to investigate the structure activity relationship of the catalysts. A possible reaction mechanism was also proposed, and it was supported by ¹H-NMR and infrared spectra (IR) analysis of the reaction solutions.

Experimental section

Materials and reagents

All chemicals used in the experiments were commercially available. 5-Hydroxymethylfurfural was sponsored by Hefei Leaf Biotech Co., Ltd. Pd/C (10 wt%) was purchased from Alfa Aesar (China) Chemicals Co., Ltd. ZrO(NO₃)₂, Fe(NO₃)₃·9H₂O, Ni(NO₃)₂·6H₂O, Ce(NO₃)₃, Se, Co(NO₃)₂·6H₂O, Pb(NO₃)₂, Bi(NO₃)₃, K₂CO₃, NaOAc, NaHCO₃, Na₂CO₃, NaOH, KOH, Et₃N, KH₂PO₄, ethyl benzoate, conc HCl, conc HNO₃ and SOCl₂ were purchased from Sinopharm Chemical Reagent Co., Ltd. *t*-BuOLi and *t*-PenONa were purchased from Merger (Shanghai) Chemical Technology Co., Ltd.

Catalyst preparation and characterization

Co(NO₃)₂·6H₂O (146 mg, 5 mmol) and Bi(NO₃)₃ (198 mg, 5 mmol) were dissolved in a mixture of 2 mL HCl and 0.2 mL conc HNO₃ while stirring at room temperature for 5 minutes and then, this solution was added to a flask containing 32 mL of deionized water and 530 mg 10 wt% Pd/C. The reaction was

maintained at 50 °C for 3 h. After 3 h, 1.2 mL of 30% NaOH solution was added to adjust the pH of the mixture to alkaline. Then, 0.4 mL of 37% formaldehyde was added, and the mixture was heated to 80 °C under Ar₂ for 16 h. After the mixture was cooled to room temperature, the catalyst was filtered, washed several times with deionized water until the pH was neutral, and dried under vacuum at 70 °C for 48 h to afford the heterogeneous catalyst (PdCoBi/C).

The catalysts were characterized by X-ray photoelectron spectroscopy (XPS) and X-ray power diffraction (XRD). X-ray photoelectron spectroscopy (XPS) data were obtained with a Thermo Scientific Escalab 250-X-ray photoelectron spectrometer equipped with a hemispherical electron analyzer and an Al K α X-ray source. All binding energies were referenced to the C 1s line at 284.7 eV. X-ray power diffraction (XRD) patterns were obtained on an X'pert (PANalytical) diffractometer with Ni-filtered Cu-K α radiation at 40 kV and 40 mA. The 2 θ range was 20°–80°. ¹H-NMR and ¹³C-NMR spectra were recorded on a Bruker Avance 400 spectrometer at ambient temperature. NMR spectra were analysed with the MestReNova software.

Typical experiment and product analysis

Twenty mol% K₂CO₃, 5 mol% Pd(10 wt% Pd/C), 5 mol% Co(NO₃)₂, 5 mol% Bi(NO₃)₃ and 2 mL MeOH were added to a Schlenk tube equipped with a stir bar. The vessel was evacuated and agitated under 1 atm O₂. After the mixture was heated to 60 °C, 0.5 mmol 5-HMF was added to the vessel, and the reaction was maintained at 60 °C under 1 atm for 14 h. After the reaction, the mixture was cooled to room temperature, SOCl₂ was added, and the mixture was stirred at room temperature for 2 h; 0.5 mmol ethyl benzoate was added as an internal standard. Then, the solution was analyzed with HPLC. The conversion of 5-HMF and the yield of products were calculated according to the following formulas:

$$\text{Conv}/\% = \frac{n_{5\text{-HMF}} - n'_{5\text{-HMF}}}{n_{5\text{-HMF}}} \times 100\%$$

$$Y/\% = \frac{n_{\text{(mole of product)}}}{n_{5\text{-HMF}}} \times 100\%$$

$n_{5\text{-HMF}}$: mol of 5-HMF before reaction; $n'_{5\text{-HMF}}$: mol of 5-HMF after reaction

Typical heterogeneous experiment and product analysis

Twenty mol% K₂CO₃, 5 mol% PdCoBi/C catalyst, and 2 mL MeOH were added to a Schlenk tube equipped with a stir bar. The vessel was evacuated and agitated under 1 atm O₂. After the mixture was heated to 60 °C, 0.5 mmol 5-HMF was added to the vessel, and the reaction was maintained at 60 °C under 1 atm O₂ for 14 h. After the reaction, the mixture was cooled to room temperature, and 0.5 mmol ethyl benzoate was added as an internal standard. Then, the solution was analyzed with HPLC. The conversion of 5-HMF and the yield of products were calculated according to the following formulas:

$$\text{Conv}/\% = \frac{n_{5\text{-HMF}} - n'_{5\text{-HMF}}}{n_{5\text{-HMF}}} \times 100\%$$

$$Y/\% = \frac{n_{\text{(mole of product)}}}{n_{5\text{-HMF}}} \times 100\%$$

$n_{5\text{-HMF}}$: mol of 5-HMF before reaction; $n'_{5\text{-HMF}}$: mol of 5-HMF after reaction

Results and discussion

The selective oxidative esterification of HMF to DMFDCA was carried out in the presence of a Pd/C catalyst, additives and alkali in methanol solvent under oxygen at atmospheric pressure (Table 1). Additives including transition metals, non-metallic elements and metal nitrate were screened initially. First, the additives of zirconyl nitrate with iron nitrate, nickel nitrate, cerium nitrate, selenium, and cobalt nitrate were probed, but they did not give satisfactory results (Table 1, entries 1–5). Then, the combinations of iron nitrate with nickel nitrate, cerium nitrate and selenium were tested, and only small amounts of oxidative esterification products were obtained (Table 1, entries 6–8). Then, cobalt nitrate was combined with selenium or metal salts (Table 1, entries 9–13), and it was found that the yield of DMFDCA increased to 42% when cobalt nitrate and bismuth nitrate were used as co-additives (Table 1, entry 10). Considering the above-mentioned reaction results, it was concluded that bismuth nitrate and cobalt nitrate have a great effect on this reaction, which might be caused by the formation of a geminal diol intermediate state during the reaction, as the bismuth atom has strong affinity for oxygen. The formation of a Bi–O bond could promote the adsorption of the substrate on the catalyst surface to form geminal diol intermediates, which was favorable for the dehy-

Table 1 The oxidative esterification of HMF with different catalysts^a

Entry	Additives A/B	Conversion ^b [%]	Yield ^b [%]		
			HMMF	FMF	DMFDCA
1	ZrO(NO ₃) ₂ /Fe(NO ₃) ₃	32	18	7	6
2	ZrO(NO ₃) ₂ /Ni(NO ₃) ₂	42	25	7	9
3	ZrO(NO ₃) ₂ /Ce(NO ₃) ₂	42	18	8	13
4	ZrO(NO ₃) ₂ /Se	39	10	4	22
5	ZrO(NO ₃) ₂ /Co(NO ₃) ₂	29	12	3	13
6	Fe(NO ₃) ₃ /Ni(NO ₃) ₂	28	10	4	14
7	Fe(NO ₃) ₃ /Ce(NO ₃) ₃	30	18	2	9
8	Fe(NO ₃) ₃ /Se	24	6	3	14
9	Co(NO ₃) ₂ /Se	42	13	4	23
10	Co(NO ₃) ₂ /Bi(NO ₃) ₃	70	25	3	42
11	Co(NO ₃) ₂ /Ce(NO ₃) ₂	48	25	10	12
12	Co(NO ₃) ₂ /Fe(NO ₃) ₃	42	17	6	16
13	Co(NO ₃) ₂ /Ni(NO ₃) ₃	39	20	4	13

^a Reaction conditions: 0.5 mmol HMF, 2.5 mol% Pd/C catalyst, 2.5 mol% A, 2.5 mol% B, 20 mol% K₂CO₃ as a base, 2 mL MeOH, under 1 atm O₂, 14 h, 60 °C. ^b The yields were determined by HPLC; ethyl benzoate was used as an internal standard. HMMF: 5-hydroxy-methyl-2-methyl-furoate. FMF: 5-formyl-2-methyl-furoate. DMFDCA: dimethyl furandicarboxylate.

drogenation of geminal diol intermediates to obtain acids by using Pd-based catalysts. Meanwhile, the formation of complexes adsorbed on the metal surface also inhibited the Pd catalyst acidosis.²⁴ It was found that the conversion of HMF and the selectivity of the products were still poor, which might be related to the reaction conditions.

To gain further insight into our reaction, a detailed analysis of the reaction conditions was carried out.

Effect of catalyst ratio

The effect of catalyst ratio on product distribution was investigated, and the results are shown in Fig. 1. No reaction occurred without the addition of Pd/C catalysts. It has been reported that the Pd/C catalyst can catalyse the conversion of alcohol to carboxylic acid, but the reaction requires an equivalent amount of base.²⁵ The yield of DMFDCA was only 30%, and the conversion of HMF was 65% with the Pd/C catalyst alone. This showed that Pd/C could catalyse the reaction but with lower yield. We did not obtain oxidative esterification products when we used single $\text{Co}(\text{NO}_3)_2$ or $\text{Bi}(\text{NO}_3)_3$ or $\text{Co}(\text{NO}_3)_2$ and $\text{Bi}(\text{NO}_3)_3$ as co-additives. The conversions of substrates were significantly improved to 94% and 92% by using the Pd/C + $\text{Bi}(\text{NO}_3)_3$ or the Pd/C + $\text{Co}(\text{NO}_3)_2$ catalyst system, respectively. This showed that the presence of Co or Bi promoted the oxidative esterification reaction.²⁶

It was found that the yield of HMMF obtained by using Pd/C + $\text{Bi}(\text{NO}_3)_3$ co-catalysts was lower than that obtained by using Pd/C + $\text{Co}(\text{NO}_3)_2$ co-catalysts, and the yield of DMFDCA was higher than those of Pd/C + $\text{Co}(\text{NO}_3)_2$ co-catalysts. This indicated that the presence of Bi was conducive not only for

the conversion of substrate, but also for the promotion of oxidative esterification.²⁷ The use of $\text{Bi}(\text{NO}_3)_3$ alone had no effect on the reaction, but it greatly increased the catalytic activity and product selectivity when combined with Pd/C + $\text{Co}(\text{NO}_3)_2$. Comparing with the experimental results in Table 1, the selectivities of products were still poor after using Pd/C + $\text{Bi}(\text{NO}_3)_3$ + $\text{Co}(\text{NO}_3)_2$ as co-catalysts, and the amounts of catalyst used were 2.5 mol%. The yield of DMFDCA increased greatly to 93%, and no FMF was observed when the amounts of catalyst used were increased to 5 mol%. On this basis, the catalyst ratio was further investigated (Fig. 1). The yields of DMFDCA were 62% and 89% when the ratios of Pd : Co : Bi were 5 : 5 : 2.5 and 5 : 5 : 7.5, respectively. The yields of DMFDCA were 63% and 87% when the ratios of Pd : Co : Bi were 5 : 2.5 : 5 and 5 : 7.5 : 5, respectively. A decrease in the amount of $\text{Co}(\text{NO}_3)_2$ or $\text{Bi}(\text{NO}_3)_3$ was not conducive to the increased yield of DMFDCA. The yield of DMFDCA significantly decreased to 53% as the proportion of Pd decreased to 2.5% (Pd : Co : Bi = 2.5 : 5 : 5). The results showed that increasing the amount of a single additive could promote the reaction. The yield of DMFDCA was similar as the ratio of Pd : Co : Bi increased from 5 : 5 : 5 to 7.5 : 7.5 : 7.5, and the optimal ratio of catalysts was 5 : 5 : 5.

Effect of alkaline species

The alkali strength and alkalinity have important effects on oxidative esterification of alcohols. The effect of bases on product distribution was investigated, and the results are shown in Fig. 2. The weak bases such as sodium acetate, sodium bicarbonate and sodium carbonate were investigated. Only 21% conversion of HMF and 7% yield of the oxidative esterification products were obtained by using sodium acetate as an alkali. The yield of DMFDCA increased to 32% by using sodium bicarbonate as the alkali; this showed that the rate of oxidation increased with an increase in alkalinity. This was

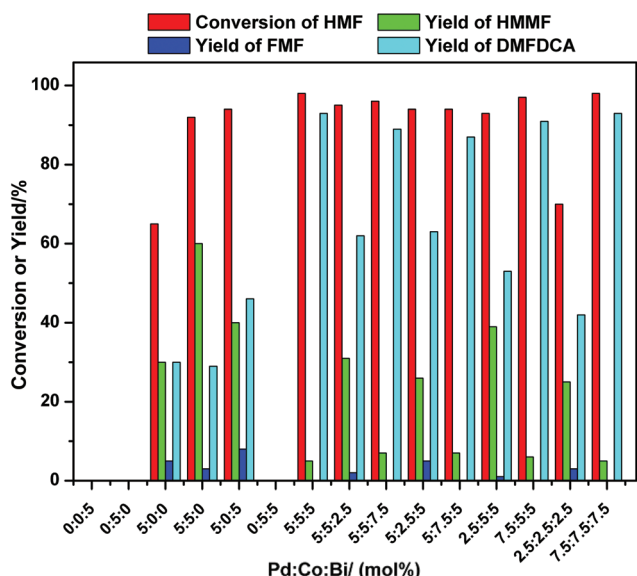


Fig. 1 Reaction conditions: 0.5 mmol HMF, 20 mol% K_2CO_3 as a base, 5% Pd/C as the catalyst, $\text{Co}(\text{NO}_3)_2$ and $\text{Bi}(\text{NO}_3)_3$ as additives, 2 mL MeOH, 1 atm O_2 , 14 h, 60 °C. The yields were determined by HPLC; ethyl benzoate was used as an internal standard. HMMF: 5-hydroxymethyl-2-methyl-furoate. FMF: 5-formyl-2-methyl-furoate. DMFDCA: dimethyl furandicarboxylate.

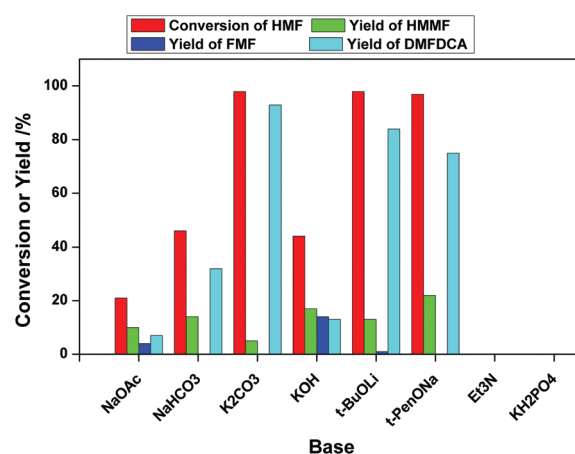


Fig. 2 Reactions conditions: 0.5 mmol HMF, 5 mol% Pd/C catalyst, 5 mol% $\text{Co}(\text{NO}_3)_2$, 5 mol% $\text{Bi}(\text{NO}_3)_3$, 20 mol% base, 2 mL MeOH, 1 atm O_2 , 14 h, 60 °C. The product yields were determined by HPLC; ethyl benzoate was used as an internal standard. HMMF: 5-hydroxymethyl-2-methyl-furoate. FMF: 5-formyl-2-methyl-furoate. DMFDCA: dimethyl furandicarboxylate.

mainly caused by the presence of alkali, which promoted the hydration of the aldehyde. The hydration of the aldehyde group to the geminal diol group was lowered with a lower reaction pH.^{24a} As a result, the strong base potassium carbonate was used, and the yield of DMFDCA significantly increased to 93%. The use of potassium hydroxide only generated small amounts of DMFDCA. This might be due to the gradual degradation of HMF (Cannizzaro reaction) under high pH.²⁸ An improved oxidative esterification yield was obtained by using the stronger organic bases (*t*-BuOLi and *t*-PenONa). The weak organic bases such as triethylamine did not significantly promote the reaction. Finally, the weakly acidic potassium dihydrogen phosphate (KH_2PO_4) was further investigated, and no product was observed. Therefore, an alkali of suitable strength was considered beneficial for the oxidative esterification of 5-HMF.

Effect of alkali amounts

The effect of the amount of alkali on product distribution was investigated, and the results are listed in Fig. 3. No conversion of HMF was observed in the absence of base, which indicated that alkali played an important role in the oxidative esterification reaction. The conversion of HMF occurred (25% of conversion) when the amount of alkali was increased to 10 mol%. Meanwhile, the yield of the oxidative esterification product DMFDCA was 15%, and the yield of the intermediates HMMF and FMF was 5% severally. This showed that the alkali promoted the oxidative reaction, but the rate of reaction was slower due to the small amount of alkali. Upon increasing the amount of alkali to 20 mol%, HMF was completely converted, and the main product obtained was DMFDCA.

The content of FMF in the mixture was always low. This might be due to the fast conversion rate of the aldehyde group in the reaction system, which led to a rapid conversion of the

intermediate FMF into DMFDCA. The conversion of HMF was complete, but the yield of the oxidative esterification products clearly decreased as the amount of alkali was increased to 50 mol%. This might be due to the fact that the aldehyde groups belonging to HMF and the intermediate FMF were disproportionated with the increasing amount of alkali, which led to reduced selectivity and yield of DMFDCA.²⁸ The yield of the oxidative esterification product DMFDCA was almost unchanged as the amount of alkali increased to 100 mol%. The yield of DMFDCA first increased and then decreased when the alkali amounts increased from 10 mol% to 100 mol%. Increasing the amount of alkali not only promoted the oxidative esterification reaction of hydroxyl and aldehyde groups, but also promoted the saponification reaction. Therefore, a suitable amount of alkali is beneficial for obtaining higher selectivity of DMFDCA, and an alkali amount of 20 mol% was found to be suitable.

Effect of reaction time

The effect of reaction time on product distribution was studied to better analyze the reaction pathway (Fig. 4). The conversion of HMF was about 80%, and HMMF and DMFDCA were produced after 1 h. The product HMMF, which was obtained from the esterification of the aldehyde group on HMF, was the major product with a yield of 53%. The yield of DMFDCA was 24%, whereas the yield of FMF was much lower. This indicated that the aldehyde group of HMF was easier to oxidize and esterify at the initial stage of reaction, which was probably due to the fact that hemiacetal is easily formed from the aldehyde group in the presence of methanol. It has been reported that the hemiacetal at the benzyl group is more easily oxidized than that of the aldehyde group to obtain an esterified product;²⁹ this might be the reason that the intermediate FMF

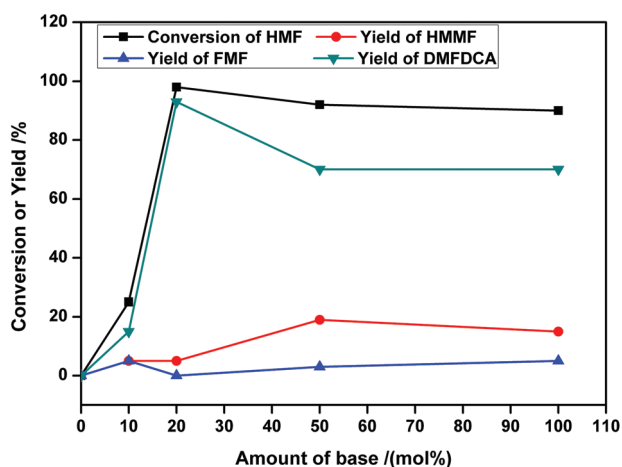


Fig. 3 Reaction conditions: 0.5 mmol HMF, 5 mol% Pd/C catalyst, 5 mol% $\text{Co}(\text{NO}_3)_2$, 5 mol% $\text{Bi}(\text{NO}_3)_3$, K_2CO_3 as a base, 2 mL MeOH, 1 atm O_2 , 14 h, 60 °C. The product yields were determined by HPLC; ethyl benzoate was used as an internal standard. HMMF: 5-hydroxymethyl-2-methyl-furoate. FMF: 5-formyl-2-methyl-furoate. DMFDCA: dimethyl furandicarboxylate.

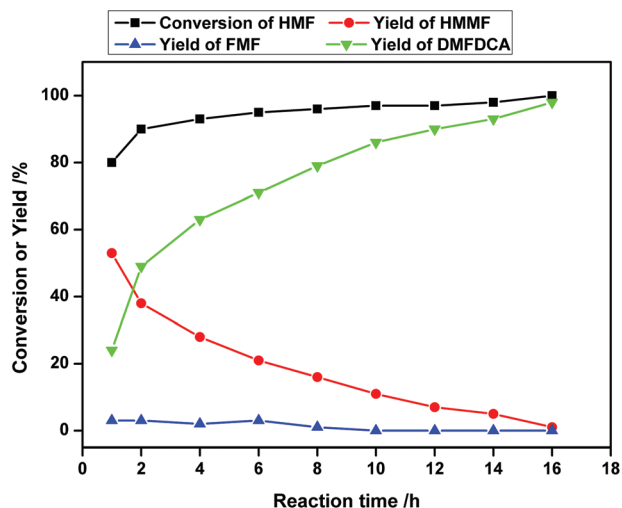


Fig. 4 Reaction conditions: 0.5 mmol HMF, 5 mol% Pd/C catalyst, 5 mol% $\text{Co}(\text{NO}_3)_2$, 5 mol% $\text{Bi}(\text{NO}_3)_3$, 20 mol% K_2CO_3 , 2 mL MeOH, 1 atm O_2 , 60 °C. The yields were determined by HPLC; ethyl benzoate was used as an internal standard. HMMF: 5-hydroxymethyl-2-methyl-furoate. FMF: 5-formyl-2-methyl-furoate. DMFDCA: dimethyl furandicarboxylate.

could hardly be detected, and it also showed that the oxidative esterification of the aldehyde groups was a fast step. The lower yield of DMFDCA also showed that the aldehyde group was not easily oxidized in the catalytic system to give a carboxyl product. With the extension of reaction time, the conversion of 5-HMF was completed, the yield of the intermediate HMMF gradually decreased and the yield of DMFDCA increased gradually. The highest yield of the oxidative esterification product reached 96% after 16 h.

Effect of heterogeneous PdCoBi/C catalysts

To study the interaction of the various components in the catalyst in a better manner, the heterogeneous catalysts were prepared according to a previously reported method,^{23c} and the effect of the catalyst ratio on product distribution was investigated; the results are listed in Table 2. The catalyst ratio had a great effect on the product distribution, according to the above-mentioned results. Similar results were obtained by using heterogeneous catalysts. The highest yield of DMFDCA was 96% when the ratio of Pd : Co : Bi = 1 : 1 : 1 (Table 2, entry 4). The yield of DMFDCA decreased to 55% when the ratio of Bi decreased from 1 to 0.5 (Table 2, entry 1), and the yield of DMFDCA decreased from 96% to 89% when the ratio of Bi increased from 1 to 1.5 (Table 2, entry 5). This showed that the ratio of Bi had a great effect on the product distribution. Similarly, the ratio of Pd and Co also affected the product yield. The yields of DMFDCA decreased to 71% and 86% when the ratios of Co decreased to 0.5 and 1.5, respectively (Table 2, entries 2 and 6). The yield of DMFDCA decreased to 68% and 82% when the ratio of Pd decreased to 0.5 and 1.5, respectively (Table 2, entries 3 and 7). Thus, it was shown that increasing the proportion of one metal could not produce the best reaction effect. The optimal ratio of Pd : Co : Bi was 1 : 1 : 1.

The yields of DMFDCA were 86% and 96% with 5 mol% and 10 mol% catalyst loading, respectively (Table 2, entries 8 and 10).

Table 2 The optimization of reaction conditions by using heterogeneous PdCoBi/C catalysts^a

Entry	Pd : Co : Bi	Conversion ^b [%]	Yield ^b [%]		
			HMMF	FMF	DMFDCA
1	1 : 1 : 0.5	91	28	7	55
2	1 : 0.5 : 1	95	19	3	71
3	0.5 : 1 : 1	94	15	10	68
4	1 : 1 : 1	99	3	0	96
5	1 : 1 : 1.5	97	5	2	89
6	1 : 1.5 : 1	97	6	3	86
7	1.5 : 1 : 1	98	9	5	82
8 ^c	1 : 1 : 1	88	1	1	86
9 ^d	1 : 1 : 1	90	62	5	20
10 ^e	1 : 1 : 1	97	19	1	76

^a Reaction conditions: 0.5 mmol HMF, 20 mol% K₂CO₃, 10 mol% PdCoBi/C catalyst, 2 mL MeOH, 1 atm O₂, 14 h, 60 °C. ^b The yields were determined by HPLC; ethyl benzoate was used as an internal standard. ^c 5 mol% PdCoBi/C catalyst. ^d No base was used. ^e Air was used instead of O₂. HMMF: 5-hydroxymethyl-2-methyl-furoate. FMF: 5-formyl-2-methyl-furoate. DMFDCA: dimethyl furandicarboxylate.

and 4). Meanwhile, the effect of alkali on product distribution was further investigated in the heterogeneous system. Under the homogeneous system, no oxidative esterification products were formed without alkali (Fig. 3). However, with heterogeneous catalysts, a 20% yield of DMFDCA was obtained without the addition of alkali (Table 2, entry 9). This showed that the interaction between the three metals was conducive for the occurrence of the oxidative esterification reaction. A 76% yield of DMFDCA was obtained by replacing oxygen with air (Table 2, entry 10).

Catalyst recirculation

The recycling experiment was carried out subsequently (Fig. 5). The experimental procedure was as follows: after the reaction, the solution was centrifuged to separate the catalyst; the catalyst was washed with methanol twice, centrifuged and used directly for the next cycle. The effect of catalyst circulation on product distribution is listed in Fig. 5. The conversion of raw materials and the yield of the oxidative esterification product gradually decreased with the increased catalyst recycling time. However, only the yield of the product HMMF gradually increased. This might be caused by the loss of the catalyst during the treatment process. After the catalyst was recycled four times, the amount of the recycled catalyst was weighed, and fresh heterogeneous catalyst was added to 10 mol%. A good catalytic activity and 91% yield of DMFDCA were obtained. Characterizations with XRD and XPS analyses of both fresh and used catalysts were conducted (Fig. S1–S4†). No significant changes in the fresh and used catalysts were found according to the analyses of the XRD spectra and XPS spectra. The catalyst also exhibited better stability.

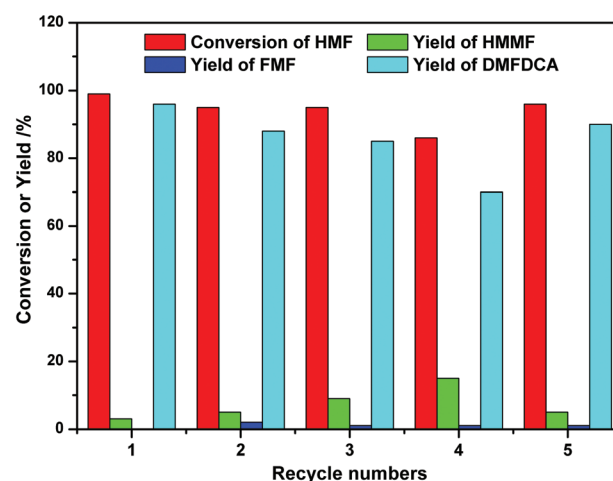


Fig. 5 Reaction conditions: 0.5 mmol HMF, 20 mol% K₂CO₃, 10 mol% PdCoBi/C (1 : 1 : 1) catalyst, 2 mL MeOH, 1 atm O₂, 14 h, 60 °C. The yields were determined by HPLC; ethyl benzoate was used as an internal standard. Number 5: fresh heterogeneous catalyst was added to 10 mol% after recycling 4 times. HMMF: 5-hydroxymethyl-2-methyl-furoate. FMF: 5-formyl-2-methyl-furoate. DMFDCA: dimethyl furandicarboxylate.

Scope of substrates

Oxidative esterifications of different alcohols in the presence of catalysts were also investigated. It was found that good yields could be obtained with aryl substrates. The substrates containing electron-donating groups (*e.g.*, methoxy and methyl) could be converted to the desired products smoothly with good to excellent isolated yields, whereas the yields of the substrates with electron-withdrawing groups (*e.g.*, nitro) were slightly decreased. With *p*-chlorobenzyl alcohol as a substrate, the oxidative esterification product was obtained with a much lower 70% yield, which might be caused by the dehalogenation reaction in the presence of a strong base. High yield was obtained with furfuryl alcohol as a substrate, indicating that this catalytic system has excellent compatibility with furan derivatives. With the fatty alcohols, such as octanol, moderate yield was obtained (Table 3).

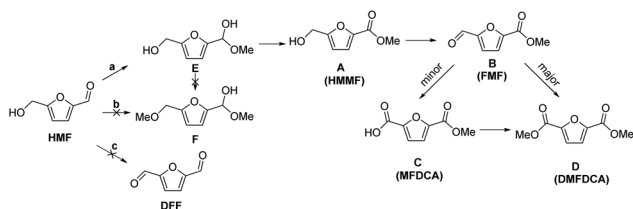
Reaction mechanism

A reaction mechanism was proposed according to a previous study of the reaction process (Scheme 2). The conversion of HMF might undergo three reaction pathways in the presence of a catalyst, methanol and oxygen: (a) the hemiacetal intermediate, **E**, was obtained through the hemiacetal reaction of the aldehyde group on HMF; (b) the etherified intermediate, **F**, was obtained through etherification of the hydroxyl group on HMF; (c) the product DFF was obtained from the oxidation of the hydroxyl group on HMF. No production of intermediates **F** and DFF was observed through monitoring of the products. This was mainly due to the fact that the etherification reaction cannot occur under alkaline conditions.

Table 3 The scope of substrates

$\text{R-OH} \xrightarrow[\text{20 mol \% K}_2\text{CO}_3, 60^\circ\text{C}, \text{1 atm O}_2, \text{MeOH, 14 h}]{\text{PdBiCo/C (10 mol\%)}} \text{R-O-C(=O)-Me}$	
	88%
	89%
	97%
	70%
	80%
	96%
	65%

Reaction conditions: 0.5 mmol HMF, 20 mol% K_2CO_3 , 10 mol% PdCoBi/C (1 : 1 : 1) catalyst, 2 mL MeOH, 1 atm O_2 , 14 h, 60 °C.



Scheme 2 Possible reaction pathways.

The main product intermediate **E** was obtained due to the fact that the acetal and hemiacetal reactions of the aldehyde group can be easily carried out in alcoholic solvents. This was also the reason why no DFF product could be detected. The intermediate **E** was converted to the intermediate HMMF more easily due to the active hemiacetal structure. With the occurrence of the oxidation reaction, the hydroxyl group in the intermediate HMMF was further oxidized to obtain the aldehyde product FMF. The intermediate FMF was mostly converted to the oxidative esterification product **D** under the catalytic system, and only a small amount of the aldehyde-based oxidized product **C** was formed. This showed that the intermediate FMF was mainly converted through hemiacetalization and reoxidation of aldehyde groups rather than by a direct oxidation process. During the reaction process, the oxidation of the hydroxyl group to the aldehyde group was the rate-limiting step. To further study the mechanism, $^1\text{H-NMR}$ and infrared spectra (IR) analyses were carried out on the reaction mixture at 0.5 h, 8 h, and 14 h. The characteristic peaks of HMMF and FMF were clearly observed according to $^1\text{H-NMR}$ analysis. The $^1\text{H-NMR}$ characteristic peaks of HMF, HMMF, FMF, and DMFDCA were successfully detected at a reaction time of 0.5 h. The characteristic peaks of HMMF and FMF significantly decreased after 8 h. When the reaction time was prolonged to 14 h, the characteristic peaks of FMF disappeared, and a very small amount of HMMF was observed. The results are listed in Fig. S5–S7 (ESI †), and the corresponding statements have been added and highlighted in the manuscript.

According to the infrared spectra (IR, ESI, Fig. S8 †), clear aldehyde peaks of FMF could be observed when the reaction was carried out for 0.5 h, indicating that the intermediate FMF was generated during the reaction. In combination with the $^1\text{H-NMR}$ analysis, it was found that the characteristic peaks of the hydroxyl group were obtained as a result of the overlap of the peaks of HMMF and HMF. When the reaction was prolonged for 14 h, only the characteristic peaks of the ester group belonging to FDCA were observed.

Conclusions

The efficient conversion of HMF to the oxidative esterification product DMFDCA was carried out by using homogeneous and heterogeneous PdCoBi/C catalysts under atmospheric oxygen. The highest yield of DMFDCA was 93% with 10% Pd/C as the catalyst, $\text{Co}(\text{NO}_3)_2$ and $\text{Bi}(\text{NO}_3)_3$ as additives, 20 mol% K_2CO_3 as the base at 1 atm O_2 , 14 h and 60 °C. The highest yield of DMFDCA was 96% with 10 mol% PdCoBi/C (1 : 1 : 1) as the catalyst, 20 mol% K_2CO_3 as the base at 1 atm O_2 , 14 h and 60 °C. The catalytic system had good compatibility according to the expansion of substrate. It was found that the reduced reaction activity in the circulation was caused by the loss during the treatment, according to the XRD spectra and XPS spectra characterizations. The oxidation of the hydroxyl group to the aldehyde group was the rate-limiting step of the reaction, according to the study of reaction mechanism.

Conflicts of interest

There are no conflicts of interest to declare.

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